Written Response Coding and Review

The research team manually coded all 2,784 responses from 174 students to 16 open-ended questions according to the operational criteria below. Coding decisions were checked against the original Chinese answers, with uncertain classifications and boundary flags retained. Each unit comprised one question version, its full scenario where relevant, and one answer. Identical answer text was coded once within the same question version when its interpretation did not depend on an unresolved cross-answer reference. Matching used the unchanged item, question wording, and answer string, without whitespace, punctuation, or Unicode normalization. This produced 836 distinct item-text units, whose codes were mapped back to all student-response records. Coding materials omitted student identity, instructional condition, class, and original score. The record retained explanatory-link, actor-relation, comparison, and error codes; error type; an exact evidence span; a short Chinese rationale; and a boundary flag.

Operational definitions

Feature	Operational coding rule
Explanatory links	0: no articulated dependency; 1: at least one explicit link between a condition, action, constraint, or consequence; 2: two connected links or an explicit interaction of conditions. Separate unconnected links remain 1.
Actor relations	1 when the answer relates at least two actors or traffic streams through competition, crossing, constraint, response, sequencing, or speed/gap coordination; otherwise 0.
Concrete comparison	1 when the answer maps a mechanism or contrasts concrete features across both situations; otherwise 0. Applied to Pre-Q8, Post-Q14, S2-Q1, S3-Q1, and S3-Q3.
Explicit error	0: no identifiable explicit erroneous or unsupported claim; 1: an identifiable claim of this kind; U: uncertain. Error type and an original-text evidence span were also retained.

Coding instructions

1. Read each stem and scene before coding. The absence of a feature is an observation, not automatically a learning deficit.

2. Do not penalize listing/identification questions for lacking a causal chain. Their explanatory-link codes can be described as auxiliary text features, not a new grade.

3. An answer saying “similar” or “more complex” alone is not a mechanism mapping.

4. Distinguish the primary merging/weaving conflict from a secondary rear-end risk; a stated crash that the stem does not report requires review or error coding, while a possible risk is not an observed event.

5. Scenario 3 does not explicitly stipulate high density. An unqualified assertion that the depicted traffic is dense is potentially unsupported; a conditional claim beginning “if density is high” is not the same claim.

6. High density does not universally imply higher speed, more lane changes, or greater collision severity. Code the actual wording, not a preferred argument.

7. A specific measure may be appropriate without an articulated explanation if the prompt only asks for a measure. Do not infer the missing reason.

8. Different critical factors or teaching priorities may be acceptable. Do not impose one model answer.

9. Keep original wording, spelling and contradictions. Do not rewrite answers to fit labels or old scores.

Concrete comparison required an explicit correspondence within the answer: a specific mechanism or feature had to be linked across the two situations. For similarity judgments, a verdict followed by features of the focal scene was treated as an implicit correspondence unless the answer explicitly connected those features across both cases. Wording such as “both” established the two-sided reference when it referred to the situations and specified a concrete shared mechanism; ambiguous references retained a boundary flag.

The original rubric could award credit for a relevant general risk factor, whereas the new error code also checked support in the stated scene. For example, high traffic density was an allowable risk indicator in the original S3-Q2 rubric, but an unqualified assertion that the scene was dense was assessed separately for support in the scenario description. The original score was retained.

Boundary flags identified uncertain error judgments, unresolved cross-answer references, or ambiguous applications of the coding criteria. Uncertain error status therefore also carried a boundary flag. Single-vehicle action order alone did not establish a relation between multiple actors. Explanatory form and correctness were coded independently. The final coded dataset retained 197 boundary-flagged distinct item-text units (615 response records), including 18 units with uncertain error status (88 records). Flagged responses remained in the primary distributions. The same-score sensitivity analysis separately excluded uncertain-error responses and all boundary-flagged responses.

Question demands and review flags

Question	Main requested response	Flagged records
Pre-Q8	Explain why weaving is more complex than merging	48
Post-Q13	Explain the roles of gaps and speed coordination	41
Post-Q14	Compare merging and weaving mechanisms	43
Post-Q15	Analyze risk sources in a short entrance–exit section	43
S1-Q1	Identify the primary conflict	12
S1-Q2	Locate conflict formation and give a reason	4
S1-Q3	Identify contributing factors	42
S1-Q4	Propose one improvement	30
S2-Q1	Explain greater complexity than standard merging	39
S2-Q2	Identify vehicle interactions	36
S2-Q3	Explain effects of high density	59
S2-Q4	Identify the most critical factor and briefly explain the choice	57
S3-Q1	Judge similarity to weaving and give a reason	38
S3-Q2	List risk indicators	6
S3-Q3	Compare conflict sources with standard merging	73
S3-Q4	Choose and explain an instructional emphasis	44

Note: Each item has 174 records. Boundary flags are a separate review field; all U-coded records also carry a boundary flag. Flagged responses remain in the primary distributions.